

Applied Bayesian Statistics in Animal & Biological Sciences

1) Introduction to Bayesian reasoning

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Introduction to myself

My positions now and previously

- PhD on (dynamic) modeling of ruminal methane production and anaerobic microbial physiology
- I also worked as a postdoc in Computational Systems Biology
- I'm currently a postdoc at the Centre for Nutrition Modelling with Jen Ellis



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I have expertise in Bayesian Statistics (in STAN and MCMCglmm)

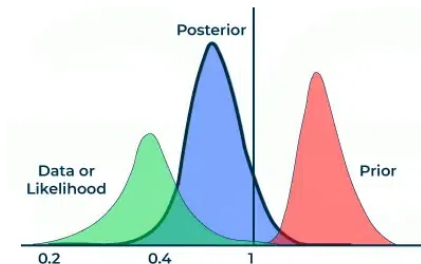
- I'm an experienced biological systems modeler, I'm not a statistician
- Disclaimer: I may not be able to exhaustively explain all concepts used in this course that I took from papers or books

Introduction of the 21 course participants

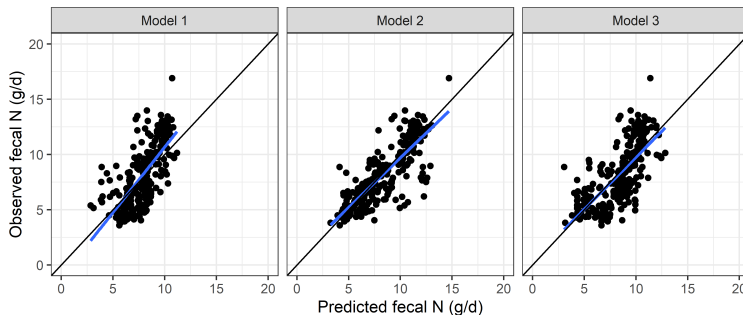
- Are you a master student, PhD candidate, postdoc or faculty member?
- What's your main research topic?
- Did you use Bayesian statistics before? In WinBUGS, MCMCglmm, BGLR, brms, STAN or PROC MCMC?
- Do you run your analyses in R, Python, C++, Julia or SAS?
- What reasons did you have to sign up for this course? Any specific analysis in mind?

Today's program

- Bayesian statistics examples in Animal science
- Frequentist statistics vs Bayesian statistics
- Bayesian reasoning
- The Bayes' rule



Application in Sheep Nutrition

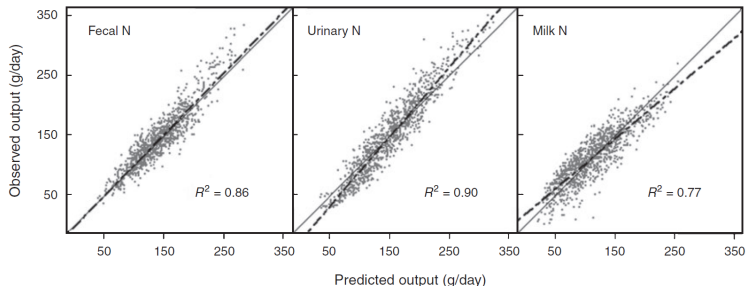


$$y_{ij} = x_{ij}^T \beta + s_i + e_{ij} \text{ with } s_i \sim N(0, \sigma_s^2), e_{ij} \sim N(0, \sigma_e^2) \quad (1)$$

Model	RMSPE
1] $-1.16 \pm 0.90 + 5.72 \pm 0.64 \times \text{DMI} + 0.196 \pm 0.026 \times \text{CP}$	24.5%
2] $-4.36 \pm 0.90 + 6.03 \pm 0.48 \times \text{DMI} + 0.109 \pm 0.021 \times \text{CP} + 0.115 \pm 0.008 \times \text{NDF}$	23.7%
3] $12.3 \pm 1.0 + 5.88 \pm 0.43 \times \text{DMI} - 0.166 \pm 0.009 \times \text{DMD} + 0.123 \pm 0.018 \times \text{CP}$	19.3%

Van Lingen et al., 2021

Application in Dairy Nutrition - Predicting N excretion



$$Y_{ijk} \sim N_3(\mu_{ijk}, \Sigma) \quad (2)$$

Y_{ijk} contains N excretion for k records, j animals and i studies, μ_{ijk} a function of a matrix of covariates X_{ijk} , Σ residual (co)-variance matrix

Reed et al., 2014

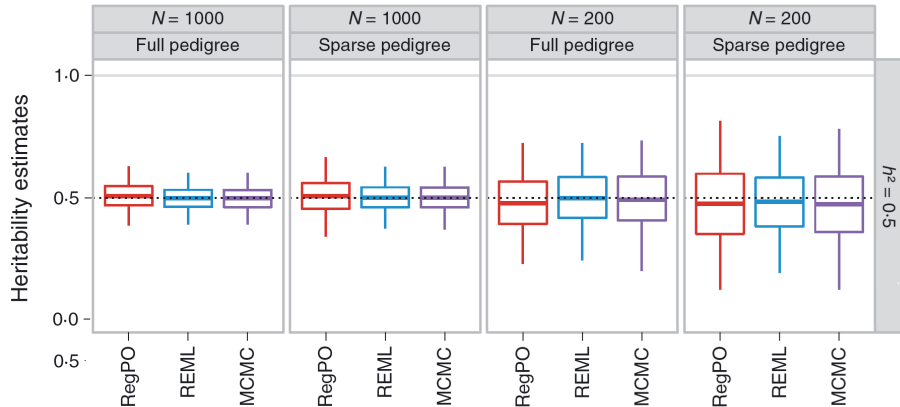
Application in Dairy Nutrition - Multivariate N prediction

$$\Sigma_1 = \begin{bmatrix} \sigma_{\text{FN}}^2 & 0 & 0 \\ 0 & \sigma_{\text{UN}}^2 & 0 \\ 0 & 0 & \sigma_{\text{MN}}^2 \end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix} \sigma_{\text{FN}}^2 & \sigma_{\text{FN}}\sigma_{\text{UN}} & \sigma_{\text{FN}}\sigma_{\text{MN}} \\ \sigma_{\text{FN}}\sigma_{\text{UN}} & \sigma_{\text{UN}}^2 & \sigma_{\text{UN}}\sigma_{\text{MN}} \\ \sigma_{\text{FN}}\sigma_{\text{MN}} & \sigma_{\text{UN}}\sigma_{\text{MN}} & \sigma_{\text{MN}}^2 \end{bmatrix}$$

- The non-zero off-diagonal case Σ_2 quantifies residual variance correlation, whereas the zero off-diagonal case Σ_1 does not

Application in Breeding & Genetics - Heritability



- Linear mixed model to assess the genetic variance component
- Similar frequentist (RegPO, REML) and Bayesian (MCMC) inference?

Frequentist P -values as a measure of evidence

P -value is not the probability H_0 is false ...

- Probability of observing the result or more extreme result under H_0
- Then a P -value is also based on "non-observed" data

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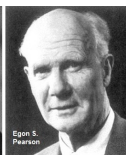
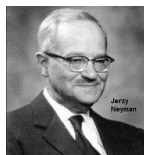
- Probability of observing the result or more extreme result under H_0
- Then a P -value is also based on "non-observed" data
- Example: CH₄ from cattle fed control and inhibitor diets:
 - $H_0 : \Delta = \mu_{\text{CH}_4, \text{control}} - \mu_{\text{CH}_4, \text{inhibitor}} = 0$
 - $\hat{\Delta} = 60$ g/d may yield $t_{\text{obs}} = 2.19$ with a P -value for the probability $t_{\text{obs}} \geq 2.19$, given that H_0 is true

Frequentist paradigms: 95% confidence interval instead?

- Fisher: strength of evidence to reject H_0 based on a P -value
- Neyman-Pearson: H_0 vs H_1 decision, minimize Type I (false positive) and Type II (false negative) errors



VS

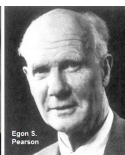
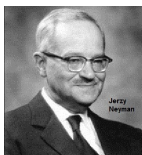


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VS



Instead, consider the Bayesian view on probability:

- There is a 95% chance that the true effect size is in the *credible interval* (more intuitive)
- One can even accompany the credible interval with the full distribution of the true effect size



Idea of Bayesian reasoning in daily life

Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)

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- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)



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Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)
- You learn the Belgian cuisine provides various delicious savoury dishes and you update your opinion (posterior knowledge)



Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water



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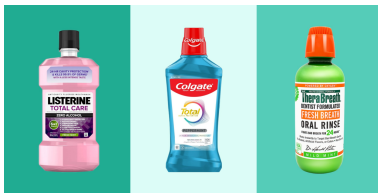
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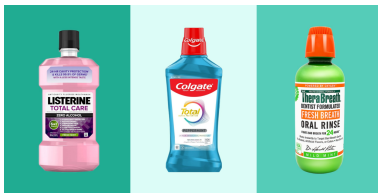
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- Result seems great, but previous products resulted in reductions of only 5% to 15%
- Use this past experience as prior knowledge and combine with the (new) data
- Update your knowledge on the efficacy of the new mouthwash



Bayesian introduction to the 21 course participants

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 - Mixed models

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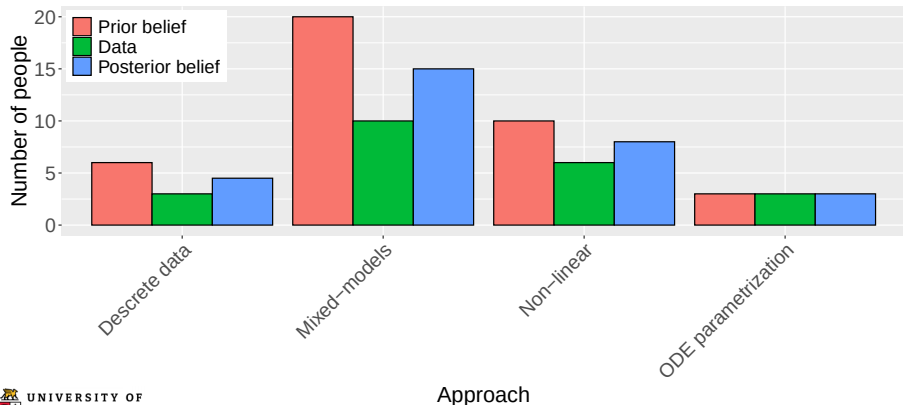
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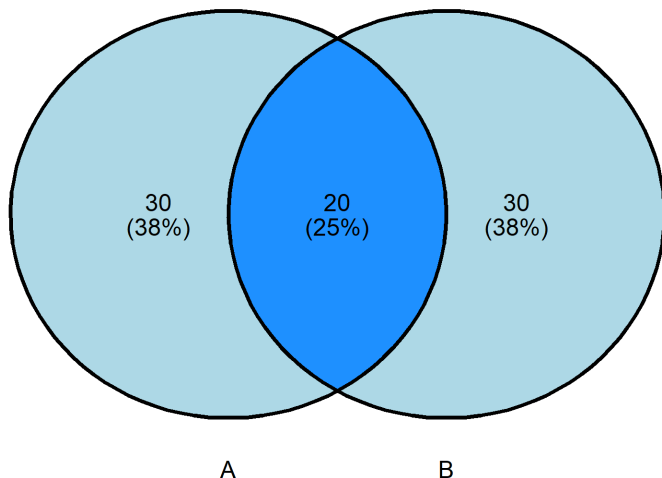
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 - Parameter estimation of an ODE model?

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Events A and B and their (joint) elementary probabilities



- Suppose: Event A is a positive diagnostic, B a diabetic patient
- Intersection or joint probability of A and B: $P(A, B)$

Probability theory and Bayes' theorem or Bayes' rule

Elementary property in probability theory:

- Marginal probabilities: $P(A)$ and $P(B)$; conditional probabilities: $P(A|B)$ and $P(B|A)$
 - $P(A|B) = \frac{P(A, B)}{P(B)}$ and $P(B|A) = \frac{P(A, B)}{P(A)}$
 - $P(A, B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

Verify/apply the Bayes' rule using Boston city hospital data

Test	Diabetic	Non-diabetic	Total
+	56	49	105
-	14	461	475
Total	70	510	580

$$\bullet P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

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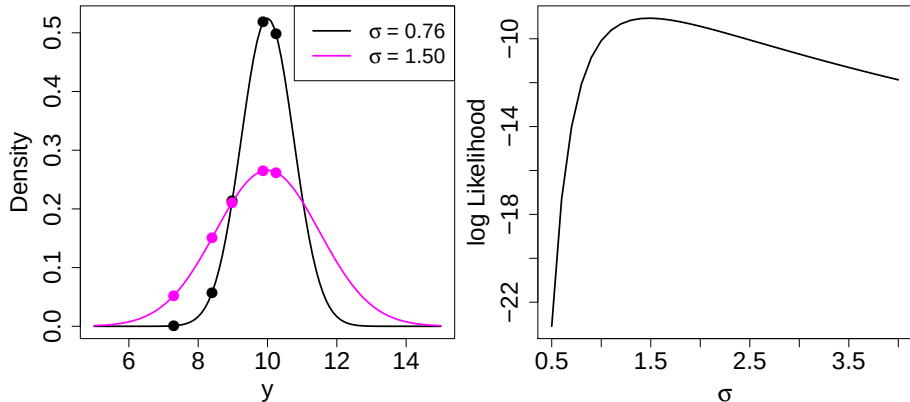
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$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)}$$
- If $P(B) = 3\%$ worldwide,
$$P(B|A) = \frac{\frac{56}{70} \cdot 0.03}{\frac{56}{70} \cdot 0.03 + \frac{49}{461} \cdot 0.97} = 0.20$$

Likelihood is used in Frequentist and Bayesian inference



- (log) Likelihood is the plausibility of model parameter values given the observed data: $L(\theta|\mathbf{y}) = P(\mathbf{y}|\theta) = \prod_{i=1}^n p(y_i|\theta)$
 - $l(\mu = 10; \sigma = 1.50|\mathbf{y}) = \log(0.052 \times 0.15 \times 0.21 \times 0.27 \times 0.26) = -9.1$
 - $l(\mu = 10; \sigma = 0.76|\mathbf{y}) = -12.9$

Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{p(\mathbf{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y}) \quad (3)$$

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This simplifies to:

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{y})}_{\text{posterior}} \propto \underbrace{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}_{\text{prior times likelihood}} \quad (4)$$

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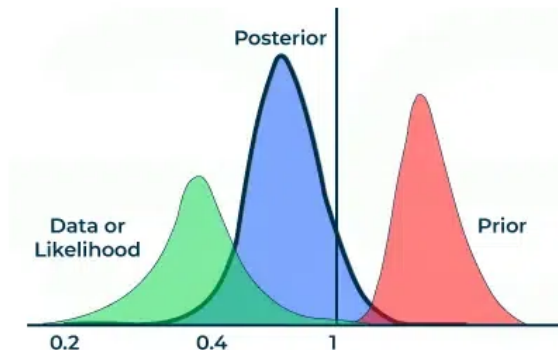
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- Parameters are considered random variables that may follow any distribution
- Optimizing $p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})$, i.e. data and prior knowledge, generates knowledge
- Powerful stochastic Markov-chain Monte Carlo algorithms generate the posterior

Prior, likelihood and posterior

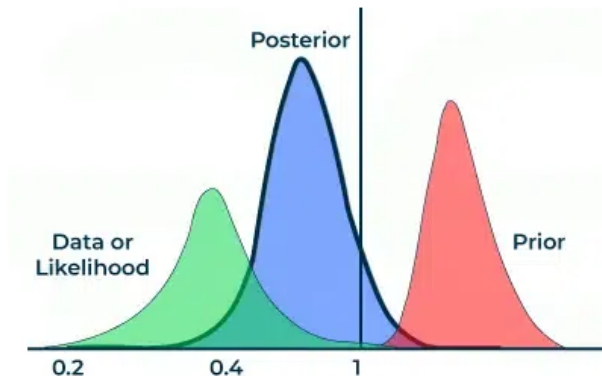


- Prior distribution: what you believed before seeing data
- Likelihood: what the data say is plausible/new evidence
- Posterior distribution: the updated belief after combining both

Reasons for choosing Bayesian statistics

- 1 Full Probability Statements Over Outcomes/Parameters - Not just one point value
- 2 Prior usage is natural
- 3 Prediction & Decision Making
- 4 MCMC is a powerful optimization procedure

Thank you for your attention!



Questions?